

Agricultural Blockchain Test Checklist

Use this sheet to test whether all seven portals, the RFID traceability flow, and the cross-portal journey are properly implemented. It also includes practical improvements so the system becomes easier for farmers, warehouses, labs, firms, government reviewers, and customers to use.

Test information

Project / Website name

Agricultural blockchain website

Website link

<https://crdtblockchain-five.vercel.app/>

Tester name

Your name

Test date

dd/mm/yyyy

Total checklist points

51

Completed points

0

Remaining points

51

Portal blocks

7

Cross-system checks

☐

Unique batch ID is created for every product batch.

Batch ID should connect farmer, warehouse, lab, firm, customer, and government views.

☐

RFID / tag number is mapped to the correct batch.

Each movement should be traceable through a tag, scan event, or batch identity record.

**Journey timeline is visible end-to-end.**

Farmer to warehouse, warehouse to lab, lab to warehouse, warehouse to firm, firm to customer.

**Status changes are visible in real time or near real time.**

Examples: submitted, stored, sample sent, passed, rejected, dispatched, sold.

**Proof documents can be uploaded and viewed.**

Photos, certificates, test reports, dispatch records, invoices, and compliance records.

**Notifications are triggered to the right stakeholders.**

Farmer, warehouse manager, lab, firm, and authority should know what action is pending.

**Rejected batch flow is defined.**

If testing fails, batch return, local sale option, reason for rejection, and history should all be visible.

**Audit trail cannot be silently altered.**

Important for blockchain trust, dispute handling, and government review.

Cross-system observations

Write notes on RFID scans, traceability flow, missing links, broken pages, duplicate records, or anything else.

**1. Farmer portal**

Farmer should be able to create a batch, upload key details, see nearby warehouses, and track the batch journey and certification result.

0%

**Add new batch option is present.**

Form should allow a farmer to create a fresh product batch.

**Batch photo upload is available.**

Farmer can upload a product or batch image.

**Farmer profile photo is visible.**

A small passport-size photo or profile panel should be shown.

**Warehouse selection is available while creating or dispatching the batch.**

Farmer can choose which warehouse the goods will be sent to.

**Nearest warehouses are shown based on location.**

This should reduce confusion and make dispatch easier.

**Batch journey is visible on the same portal.**

Farmer should see every stage after submission.

**Certification result is shown clearly.**

Passed, failed, pending, and certificate-view action should be easy to understand.

**Rejected batch return message is shown to the farmer.**

If the batch fails testing, the return reason and next step should appear.

Farmer portal notes

Check usability, form validation, map/warehouse listing, batch tracking clarity, and farmer-facing alerts.



2. Warehouse portal

Warehouse records should clearly identify the facility, the manager, the receiving activity, and the quantity of goods being stored.

0%

**Warehouse address is displayed.**

Full address and contact context should be visible.

☐ **Warehouse manager name is displayed.**

User should know who is responsible for the warehouse.

☐ **Receiver / handler of incoming goods is recorded.**

The person receiving storage goods should be identifiable.

☐ **Stored quantity is recorded.**

Actual quantity, unit, and batch mapping should be visible.

☐ **Warehouse-in and warehouse-out events are logged.**

Movement in and movement out should appear in the trace history.

☐ **Storage condition details are visible.**

Temperature, cleanliness, and storage suitability are especially important for honey.

Warehouse portal notes

Check receiving logs, stock display, storage records, warehouse assignments, and tag scan records.



3. Quality testing lab portal

The lab should receive a sample from the warehouse, test it on defined parameters, compare values to ideal ranges, and issue or deny a certificate.

0%

☐ **Nearest quality lab linkage exists from warehouse flow.**

Sample should move from warehouse to the nearest lab logically.

☐ **Testing parameter list is visible.**

The website should show the matrices or testing criteria already defined for the product.

☐ **Ideal range is shown near each testing field.**

This can be shown in brackets, helper text, or slider guidance.

☐

Lab can enter actual test values.

Each parameter should accept recorded lab values.

☐

Pass / reject logic works on the basis of ideal range.

Out-of-range results should trigger rejection or failure handling.

☐

Quality certificate can be generated for passed batches.

Certificate should be viewable or downloadable.

☐

Rejected batches do not get a certificate.

The system should prevent accidental approval.

☐

Lab result flows back to warehouse and farmer portal.

Certification status should update all dependent portals.

Lab portal notes

Check input fields, value validation, pass/fail automation, certificate visibility, and return of results to linked portals.



4. Quality testing officer portal

The officer should be able to inspect storage quality, hygiene, and product quality conditions, not just lab values.

0%

☐

Officer can review testing ranges and quality thresholds.

Important values should be visible without confusion.

☐

Officer can inspect warehouse cleanliness.

Portal should support recording hygiene and storage inspection observations.

**Storage conditions for the product can be checked.**

For honey, storage conditions must match product requirements.

**Officer can record full inspection findings.**

Observations, remarks, and supporting proof should be saved.

**Officer can validate whether product quality is acceptable.**

Portal should support inspection outcome capture.

**Inspection results affect downstream workflow where required.**

For example, holding a batch, escalating it, or approving further movement.

Quality officer portal notes

Check inspection reporting, condition records, hygiene evidence, remarks section, and workflow controls.



5. Firm portal

The firm should be able to re-test if needed, upload its own result, and then release the verified product to market.

0%

**Firm can review batch details and previous test reports.**

Firm should see warehouse and lab history before further action.

**Firm can mark whether re-testing is done.**

Retest status should be explicit, not assumed.

**Retest result upload is mandatory if re-testing is done.**

This should be visible to customer and other stakeholders.

**Firm can approve dispatch to market.**

Once verified, the batch can move for customer sale.

**Customer-facing traceability still shows firm-stage actions.**

Transparency should continue after firm handling.

Firm portal notes

Check retest toggle, report upload, market dispatch state, and visibility to customer timeline.



6. Government authority portal

Authority users should be able to review the full chain, detect issues, and intervene where fraud, alteration, or process gaps are found.

0%

**Authority can view the full end-to-end process.**

Nothing important should be hidden from the monitoring role.

**Authority can inspect records stage by stage.**

Portal should support review of farmer, warehouse, lab, firm, and customer-side history.

**Authority can detect suspicious alteration or inconsistency.**

Any tampering, mismatch, or missing step should be reviewable.

**Authority can flag, hold, or scrap problematic entries.**

Action should be recorded with reason and timestamp.

**Authority actions appear in the audit trail.**

Oversight activity should also be transparent and reviewable.

Government portal notes

Check monitoring power, intervention controls, audit review, and visibility across all stages.



7. Customer portal

The customer mainly needs a clear view of authenticity, product journey, and quality proof, without unnecessary complexity.

0%

☐

Customer can view the complete blockchain journey.

Timeline should be simple, visual, and trustworthy.

☐

Customer can confirm authenticity of the product.

The portal should make it obvious that the honey is verified and genuine.

☐

Customer can see test certificate / report.

Quality evidence should be accessible without confusing navigation.

☐

Farmer or source identity is visible in a suitable format.

This builds trust and supports traceability.

☐

Customer view is easy to understand on mobile.

Most customers will likely access this through a QR or scan flow.

Customer portal notes

Check scan journey, timeline readability, trust markers, certificate access, and mobile usability.



Recommended improvements

Scan-first workflow

Add RFID and QR scan buttons on every movement screen so users do not have to search for a batch manually each time.

Status colors

Use simple status chips like pending, in transit, under test, passed, rejected, and returned for faster understanding.

Mandatory reason fields

Whenever a batch is rejected, held, or edited, require a reason, user name, timestamp, and proof attachment.

Geo support

Show nearest warehouses and labs on a small map with distance, estimated travel time, and available capacity.

Role dashboards

Each portal should show pending tasks first, for example batches waiting for pickup, testing, approval, or return.

Mobile forms

Make forms short and step-based so farmers and field officers can complete them easily on a phone.

Certificate preview

Show a quick preview of the latest quality certificate before downloading the full document.

Dispute handling

Add a flag or dispute option where a stakeholder can report mismatch in weight, scan, quantity, or test result.

Final remarks**Overall website review**

Write the final result of your test run here: what is complete, what is missing, and what needs urgent fixing.



Tip: Use this checklist during one full batch simulation from farmer creation to customer view. That will reveal missing links faster than checking portals separately.